



BALL STATE
UNIVERSITY

Practical Assignment II

Analysis of Variance

Quinton Quagliano, M.S., C.S.P

Department of Educational Psychology

Table of Contents

1	Pre-assignment Reminders	2
2	Context	2
3	Instructions	3
3.1	Knowledge Checks (6 pts)	3
3.2	SPSS Applied Work (14 pts)	3

1 Pre-assignment Reminders

- Prior to attempting this homework, I strongly recommend that you have completed all other work through module 6, including:
 - Likely the most useful resource will be the video I specifically record for this assignment (near the assignment portal on Canvas)
 - Having watched all lectures thoroughly and taken notes (optionally with the guided notes)
 - Reading all assigned sections of the book, and completing practice examples as helpful and useful
 - Completing relevant lecture check-ins, and using answer key for review
 - Completing all quizzes, and reviewing correct answers at the end of the week
- This practical assignment will be graded for accuracy, please give yourself enough time to complete your best work
 - You may use your notes, book, SPSS software, provided data, and lectures to aid you in completing this
 - Where possible, include more detail to ensure you fully explain your rationale for a question
 - You should include ALL relevant SPSS output to support your work, please make clear where in the output you get your answers from
- This practical assignment is cumulative to the modules and practical assignment(s) prior, please review your previous work and notes to help you accomplish this work
- All syllabus and university policies on academic integrity, plagiarism, and other forms of misconduct apply to this assignment. Please review them if you are unfamiliar

2 Context

The following information will be useful for answer some questions on the homework, please ensure you read and understand the following context

You have gathered basic information about several high school students, and would now like to examine several continuous outcomes based upon several grouping factors.

Your goal is to use your skills in SPSS and growing statistical knowledge to analyze the distributions and descriptives of the variables, and then apply inferential parametric and non-parametric analyses.

Please use the provided dataset on Canvas, edu_data_2.sav, to complete this assignment.

3 Instructions

3.1 Knowledge Checks (6 pts)

1. Describe the 3 assumptions of the parametric independent samples t-test (2 pts)
2. Describe the 3 assumptions of the parametric one-way ANOVA (2 pts)
3. Describe 1 visual and 1 statistical way to assess for normality in data (1 pts)
4. Describe 1 statistical way to assess for homogeneity of variances in data (1 pts)

3.2 SPSS Applied Work (14 pts)

For each of the following questions, please provide any and all relevant SPSS output (or syntax when indicated). Follow the directions IN ORDER

1. Use the “Explore” descriptives function to show the mean, median, standard deviation, and range of attendance rate GROUPED BY internet access. This should also include histograms with a normal curve overplayed, for each level of the internet access variable. (2 pts)
2. For the above procedure, explain why or why not you feel there is evidence of normality in the attendance rate variable when separated by internet access. Provide concrete number or reference visuals to back up your claim. (1 pts)
3. Use the “Explore” descriptives function to show the mean, median, standard deviation, and range of attendance rate GROUPED BY school type. This should also include histograms with a normal curve overplayed, for each level of the school type variable. (2 pts)
4. For the above procedure, explain why or why not you feel there is evidence of normality in the attendance rate variable when separated by school type. Provide concrete number or reference visuals to back up your claim. (1 pts)
5. Run a parametric independent-samples t-test on attendance rate between the “Yes” and “No” levels of the internet access variable. Report the test statistic, p-value, and whether Levene’s test suggest homogeneity of variances. Describe whether there are significant differences or not (2 pts)
6. Run a Mann-Whitney U test on attendance rate between the “Yes” and “No” levels of the internet access variable. Report the test statistic and p-value. Describe whether there are significant differences or not (1 pts)
7. Run a parametric One-way ANOVA on attendance rate between the 3 levels of the school type variable. Report the test statistic, p-value, and whether Levene’s test

- suggest homogeneity of variances. Describe whether there are significant differences or not (2 pts)
8. Choose one type of post-hoc test to analyze contrasts between the groups, explain what they tell you about the differences between each group, and provide evidence from your SPSS output to support this (1 pts)
 9. Run a Kruskal-Wallis test on attendance rate between the 3 levels of the school type variable. Report the test statistic and p-value. Describe whether there are significant differences or not (1 pts)
 10. Using the post-hoc test, analyze contrasts between the groups, explain what they tell you about the differences between each group, and provide evidence from your SPSS output to support this (1 pts)